



ENGR 1166 Portfolio

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Table Of Contents

Iterations

Individuals should be well-versed in the basic ideas, theories, and models that form the basis of their field of study. This entails being aware of important terms, ideas, and connections between various ideas. People ought to be able to use their theoretical knowledge to develop workable solutions for problems or opportunities in their line of work.

Communication

To ensure that team members comprehend one another's viewpoints, share information efficiently, and coordinate their activities, clear and open communication is essential. This entails giving constructive criticism, speaking clearly, and actively listening. Establishing a collaborative atmosphere requires acknowledging and appreciating the distinct abilities, expertise, and viewpoints that every team member contributes.

Teamwork

Evaluating the existing design and determining its advantages, disadvantages, and potential improvements. modifying the design in light of the analysis carried out in the preceding phase. This could be enhancing functionality, perfecting the look, or maximizing efficiency to solve problems found and meet project objectives.

Ethics

Incorporating moral issues into design at every level, from conception to execution and beyond. This guarantees that the systems, solutions, or products being designed respect human rights, give priority to ethical values, and take into account the wider societal consequences of their use. Designers ought to put the rights and welfare of those impacted by their designs first.



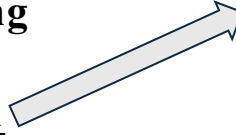
Jada Fox

CRB Air-Filter

Class Name: Engr 1166

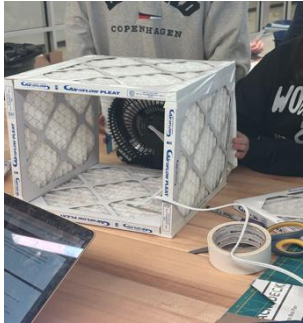
Major: Biomedical Engineering

Learning Outcome 1



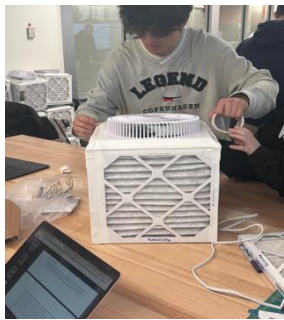
The objective of this assignment was to test the previously built CRB ventilation boxes and filter the air quality in rooms that could have air pollution in them. To test this, you could use a humidifier to ‘pollute’ an enclosed room and then graph the air quality measurements were recorded after time to ensure that the CRB had done what it was designed to do.

Understanding Concepts



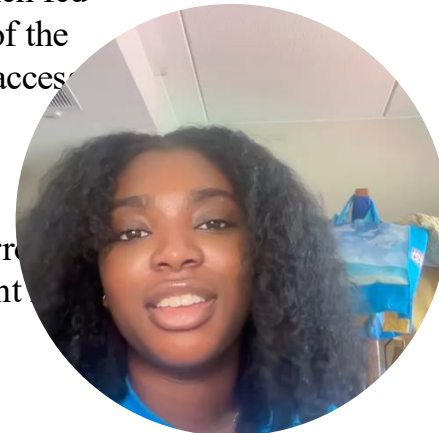
We began our project by placing four of the filtration systems into an ‘L’ formation, this was to create the 90-degree angle needed to form a cube foundation for the fan to be placed. This part is essential to the entirety of the rest of the build. We then used duct tape to secure the placement of the filters and prevent the leakage of air. Without a stable and angled structure the entire filtration system would have collapsed.

Once we created the foundation, we began the next vital step which was to create support for the fan inside of the filtration system. We carved the cardboard box packaging from the fan into a sort of headboard support system. This provides stability for the fan itself and allows us to use tape to connect the fan to the rest of the filtration system.



After creating the cardboard support system, we taped the cardboard onto the fan, and then taped the combination of cardboard and fan to the cubic filter foundation. We then fed the wire of the fan into the bottom of the filtration system. This puts the separated parts of the project together with duct tape placed around the corners of the filters and allows easier access to a power source.

After successfully building the air filtration system we went and reapplied duct tape to all connections of the filtration system and then ran the fan. We did not find any errors or faults in the fan itself or the set up of our filters. Our overall build and function of the ventilation system was successful.



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CRB Air -Filter

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Learning Outcome 2

Construction



The goal of this lab study is to create inexpensive air filters that will raise the standard of the surrounding air. This is important for those with respiratory conditions like allergies or asthma, as well as for maintaining comfortable and healthful indoor environments in homes, offices, and other areas.

Our concept began with four filtering systems arranged into a "L" shape to create the necessary 90-degree angle to create a cube base for the fan. This part is the only one that affects the rest of the build. Next, we used duct tape to make sure the filters were positioned firmly and to seal up air leaks. As soon as we completed the foundation, we began the next vital step: supporting the fan inside the filtration system. Using the cardboard box packaging that came with the fan, we made a sort of headboard support system. This allows us to attach the fan to the in addition to providing stability to the fan itself.

After building the cardboard support structure, we taped the cardboard to the fan and the cardboard and fan together to the cubic filter foundation. Next, the fan wire was put into the bottom of the filtering system. This unites the project's split components with duct tape across the corners of the filters and makes it simple to access a power supply. After the air filtration system was properly constructed, we tested the fan and reapplied duct tape to all the connections. We were unable to find any errors or defects in the filter itself or the fan setup.

The vent's overall design and functionality were both excellent. To assess our completed air filtration system, we plugged the fan into a power outlet to verify the taping of our structure and the filters themselves. We enhanced the filtration system by running duct tape across the wiring at the rear of the fan to make sure there was no air leakage

The biggest positive of this filter is that it is an easy way to improve air quality in places that do not meet modern standards.



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CRB-Air filter

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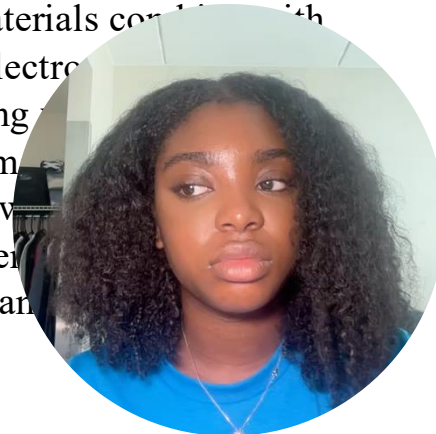
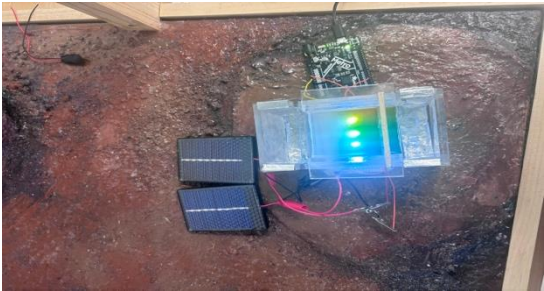
Learning Outcome 3

As a team, we decided to build the gel electrophoresis station as our model. Our team plans to work efficiently within interdisciplinary teams and Iteratively design, build and improve a device or a process to meet a specified need within given constraints.

Our team will use the unanimous decision rights model as it is the best option to ensure everyone is heard and respected. If necessary to move forward with the project we will fall back on the democratic model, but we believe to ensure no one feels slighted, the unanimous decision is the best model.

Our team will handle instances such as the ones described above by ensuring everyone is heard and understood. If someone seems to have a lack of participation, the best way to address it would be to make the other team members' opinions clear and specific ways for the team member to improve. It is inevitable that different opinions will arise, but the important matter to keep in mind is to respect and understand everyone's side.

As a mechanical engineering major, it is important that I know how different materials combine with physical properties to create the best gel electrophoresis station. I am also interested in electronics and robotics, so I will take charge of the LED and solar panel wiring. I believe that I am strong and I aim to become more aware of how different materials affect designs as part of my mechanical engineering. As a Biomedical engineer, I will work on constructing the base of the electrophoresis tray as well as the gel we are using to make sure the DNA fragments float across the gel. Alice: As a chemical engineer, I will focus on the components of the gel and samples as well as construction of the base tank.



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CRB Air -Filter

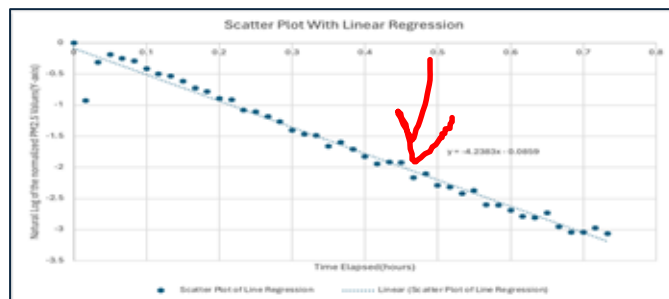
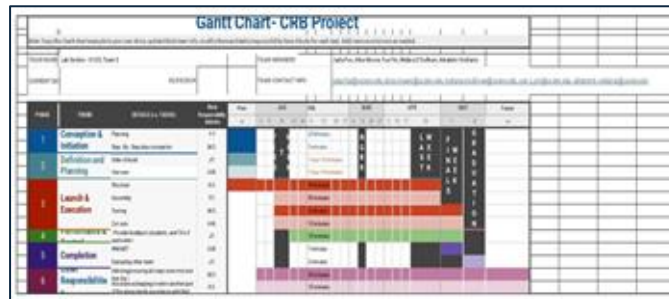
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Learning Outcome 4

Graphing and Data

ACH (h^{-1})	4.2383 h^{-1}
CADR (m^3/hr)	44.078 m^3/hr
Time you turned humidifier ON	9:21 a.m.
Time you turned humidifier OFF	9:26 a.m.
Time you turned CRB ON	9:26 a.m.



The objective of this was to test the previously built CRB ventilation boxes to test the air quality in rooms that could have air pollution in them. Using a humidifier to ‘pollute’ an enclosed room, the CRB was run and the air quality measurements were recorded after time to ensure that the CRB had done what it was designed to do.

The faster that the fan will spin, the faster the rate of air filtering through will increase. This means that as the ACH increases, so will the CADR. When the fan of the CRB is on a lower setting, the air does not filter quickly, and so the rate of ACH and CADR will be lower than if it was on a higher setting.

We used a Gantt chart to organize the planning and preparation for filter. The planning aspect of this project took about 20 minutes to complete. While building the duct it took about 1 hour and 10 minutes as we had to redesign the duct in order to make sure the arrows of the filter face one another. Testing the duct took about 5 minutes because we routinely checked the filters and tape durability. Refining the CRB took about 2 minutes because our only critique was to tape the wiring.

The data has a linear regression, as demonstrated with the Excel Plot below. If the data was not linear, like a negative exponential graph, it would indicate that a longer operation time would not significantly change the particle concentration.



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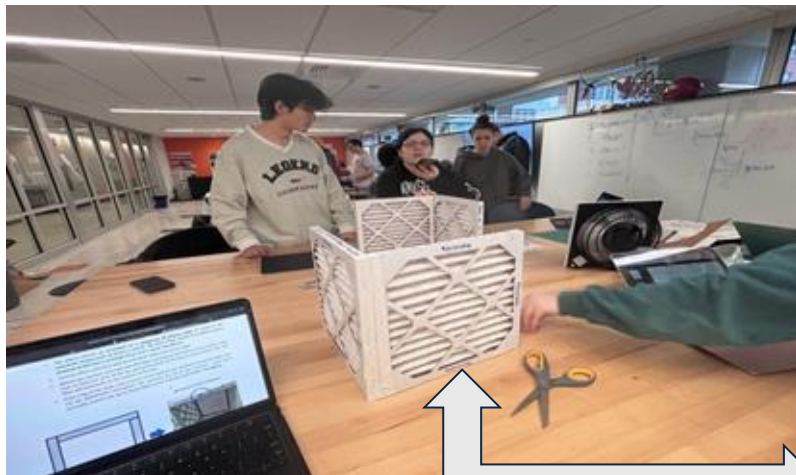
Learning Outcome 5

The lab aims to create affordable air filters to enhance indoor air quality, benefiting those with respiratory issues like allergies or asthma. This solution offers a simple way to improve air quality in non-compliant areas. Constructing the C-R box requires 5 filters and a central fan support.



Having this do-it-yourself air filter is crucial as it can enhance air quality and reduce health hazards while remaining quite economical. Handmade C-R Boxes are incredibly adaptable and individualized. To improve sleep, you can relocate them to focus on certain areas where the quality of the air may be worse, such as next to a kitchen, in a dusty or fume-filled workplace, or in a bedroom. This flexibility is especially useful if your housing situation is transient. It is a fantastic and reasonably priced way to raise one's standard of living. The Veterans Association will get this C-R box as a donation.

An ethical way to improve the filter is to make the sides collapsible for easier changing without dismantling the entire filter. The z-shaped folds allow the sides to collapse and be pulled through the opening. The metal wiring forces the filter to cover the opening, ensuring a proper fit with a larger surface area.



The design is predominantly held together by duct tape, which helps to reduce costs but may compromise stability. One solution could be to use glue in strategic areas, such as the creases between filters, to enhance stability. Adding protective covers, like cardboard, to the filters could also prevent damage and improve the overall durability of the design.

